Contents

Supporting Information: Human-AI Collaborative Build Dialogue **1**
1D CO2 Electroreduction Catalyst Layer Model . . . 1
Preface . . . 1
Sources and Completeness . . . 1
1. Model Overview . . . 1
2. Phase 1: Implementation (March 25) . . . 2
3. Phase 2: Debugging (March 25) . . . 2
Bug 1: Filename Parse Off-by-One . . . 2
Bug 2: O(n^2) Network I/O . . . 2
Bug 3: Wrong Sweep Order . . . 2
Bug 4: Neumann BC Sign Error (Critical) . . . 2
Bug 5: FD Jacobian Dead Zone . . . 3
4. Phase 3: Stage 3 Convergence (March 25–26) . . . 3
5. Phase 4: Stage 4 and Plotting (March 26) . . . 3
Stage 4 Completion . . . 3
Visualization Request . . . 3
Course Correction: CO vs CO2 . . . 3
Course Correction: Undo Edits . . . 4
6. Phase 5: Physicality Checks (March 26) . . . 4
Human Challenges AI Interpretation . . . 4
Human Identifies Real Problems . . . 4
7. Phase 6: Code Audit (March 26–27) . . . 5
Findings . . . 5
8. Phase 7: Implementing Fixes (March 27) . . . 5
Pressure Convention Discussion . . . 6
9. Phase 8: Parameter Fitting (March 27–28) . . . 6
Results . . . 6
Comparison at Experimental Voltages (Post-Fit) . . . 6
10. Session Recovery . . . 7
11. Reflections on Human-AI Collaboration . . . 7
Division of Labor . . . 7
Patterns Observed . . . 7
12. Summary of All Changes . . . 8
Bugs Fixed (5) . . . 8
Code Audit Fixes (3) . . . 8
Parameter Calibration . . . 8
Visualization . . . 8
13. File Inventory . . . 8

Supporting Information: Human-AI Collaborative Build Dialogue

1D CO2 Electroreduction Catalyst Layer Model

AI System: Claude Opus 4.6 (1M context), Anthropic **Interface:** Claude Code CLI (VS Code extension) **Programming Languages:** Julia (solver), Python (plotting/analysis) **Dates:** March 25–28, 2026

Preface

This document records the collaborative process between a human electrochemistry researcher and an AI assistant (Claude, Opus 4.6) to implement, debug, validate, and calibrate a 1D finite-volume model of CO2 electroreduction in a gas-diffusion electrode. It is intended as supporting information for publication, documenting both the technical outcomes and the nature of human-AI interaction during scientific software development.

The collaboration spanned four days and multiple Claude Code sessions. The human provided domain expertise, physical intuition, and quality control; the AI provided code generation, systematic debugging, and computational labor. The dialogue below captures key decision points, course corrections, and the iterative build-test-fix cycle that characterized the process.

Sources and Completeness

This document draws from three sources:

1. **Project memory** – persistent notes saved by the AI across sessions
2. **Recovered transcript** – a partial transcript from a session that crashed mid-work (March 26–27)
3. **Live session** – the final session (March 27–28)

Sections covering the initial implementation and early debugging (March 25) are reconstructed from memory and marked accordingly. All quoted dialogue is verbatim from transcripts.

---

1. Model Overview

The model simulates coupled gas-phase and liquid-phase transport through a gas diffusion layer (GDL, 325 um) and catalyst layer (CL, 10 um), with Marcus-Hush-Chidsey (MHC) kinetics for CO2 reduction and Butler-Volmer kinetics for hydrogen evolution. The implementation follows a staged approach, with each stage adding physical complexity:

Stage	Physics Added	DOFs
1	Liquid diffusion + buffer + kinetics, Dirichlet BCs	250
2	Neumann (Sherwood) BC at electrolyte side	250
3	Stefan-Maxwell gas transport + Darcy pressure, coupled to liquid	570
4	Pressure diagnostics (post-processing of Stage 3)	570

Each stage sweeps 734 voltage points from -0.3 to -2.5 V vs RHE, warm-starting each voltage from the previous converged solution. Experimental reference: Lees et al. 2024, 50 C, 1M KHCO3, Ag catalyst.

---

2. Phase 1: Implementation (March 25)

[Reconstructed from memory]

The human provided an implementation guide document (`CO2R_BULK_MODEL_IMPLEMENTATION_GUIDE.md`) specifying the governing equations, boundary conditions, discretization, and all physical parameters. The AI (Claude) translated this specification into a Julia codebase of 13 source files.

The key architectural decisions made collaboratively were:

- Log-concentration DOFs (`u = ln(c)`) for liquid species, avoiding negative concentrations
- Cell-centered finite-volume scheme with graded GDL mesh (geometric ratio q=1.3)

- Dense finite-difference Jacobian with row scaling (appropriate for ~570 DOFs)
- Binary `.bin` cache files for each voltage, enabling restartability on a network drive
- Predictor-corrector (PAC) continuation for voltage sweeping

The staged pipeline was designed so that each stage could be validated independently before adding the next layer of physics.

---

3. Phase 2: Debugging (March 25)

[Reconstructed from memory]

The first run attempts revealed five bugs. The debugging process illustrates how human physical intuition and AI systematic analysis complemented each other.

Bug 1: Filename Parse Off-by-One

All stage files used `f[4:end-4]` to extract voltage from filenames like `s1_V-0.300000.bin`. The prefix `s1_V` is 4 characters, so Julia's 1-based indexing requires index 5. A straightforward off-by-one caught during the first failed run.

Bug 2: O(n^2) Network I/O

The voltage sweep loops called `readdir()` or `isfile()` inside every iteration – 734 iterations x 734 cached files on a Box-mounted network drive. The AI identified this as a performance issue and fixed it by pre-building a `Dict`/`Set` of existing files before the loop, reducing ~540,000 filesystem operations to one.

Bug 3: Wrong Sweep Order

Stages 2 and 3 swept voltages from most negative (-2.5 V) to least negative (-0.3 V). Starting from the hardest case with no good warm-start caused widespread Newton failures. The fix: sweep easy-to-hard (-0.3 to -2.5 V) and chain each converged solution forward.

Bug 4: Neumann BC Sign Error (Critical)

This was the most consequential bug. All 734 Stage 2 voltages failed with residual ~1e4.

The AI's debugging approach was systematic: first examining which DOFs had singular Jacobian columns (DOF 250 = last CO2), then checking the initial residual of Stage 2 using a Stage 1 warm-start (9706 vs the expected ~0), and finally tracing the discrepancy to a sign convention mismatch. The Sherwood-Reynolds BC was coded as `k_MT * (c_eq - c)` but the correct sign (matching the `diffusion_flux` convention where positive = rightward/leaving cell) is `k_MT * (c - c_eq)`. The wrong sign made mass transfer destabilizing rather than restoring.

Bug 5: FD Jacobian Dead Zone

At `u = ln(c) = -50` (the hard clamp), forward-difference perturbation `u + 5e-6` still evaluates to -50 after clamping, producing a zero Jacobian column and a `SingularException`. Fixed by ensuring perturbations escape the clamp boundary.

---

4. Phase 3: Stage 3 Convergence (March 25–26)

[Partially reconstructed; transcript begins mid-phase]

After bugs 1–5 were fixed, Stages 1 and 2 completed (734/734). Stage 3 introduced coupled gas transport and proved more challenging.

Runs 1–4 converged for the first ~300 voltages (-0.3 to -1.13 V) but hit `SingularException(357)` at every subsequent voltage. The DOF corresponded to a gas mass fraction clamped to exactly zero – the same class of dead-zone bug as Bug 5, but for gas DOFs.

The fix: Add a lower bound `max(u_gas[k], 1e-15)` in gas DOF extraction.

Run 5 converged all 734 voltages. From the recovered transcript, the AI monitored progress in real-time:

> **AI:** 507/734 at 12:18 – through V=-1.632 V with zero failures. Rate has accelerated to ~30/min. At this rate, ~7–8 more minutes to complete all 734 voltages.
>
> **AI:** 734/734 – Stage 3 complete! All solutions converged through V=-2.499 V RHE with zero failures.

---

5. Phase 4: Stage 4 and Plotting (March 26)

Stage 4 Completion

Stage 4 copies Stage 3 solutions and computes Darcy pressure diagnostics – no Newton solving, only I/O. It completed 734/734 in approximately 30 minutes (limited by Box network drive speed).

Visualization Request

The human made a detailed, specific request for visualization:

> **Human:** Add plots in the stage 1 and 2 outputs for profiles of OH-, CO2(aq), and pH across the CL. Add plots in the output of stage 3 of profiles of CO2 and H2 mass fractions and deltaP across the modeling domain, and profiles of OH-, CO2(aq), and pH across the CL. For stage 4, add a 9 panel plot of liquid species/rxns in the CL (pH, CO2(aq), HCO3, CO3, H+, OH-, Cs+, Rco2r, Rher) and a 4 panel plot of deltaP and the partial pressures of CO2, CO, and H2 across the modeling domain. All the aforementioned plots should have x on the x axis and different data series for each V. Then add one more plot in stage 4 that is CO2 utilization as a function of V.

The AI implemented all requested plots in a single 582-line Python script, including binary file reading, mesh reconstruction, kinetics re-implementation (for R_CO2R and R_HER profiles), and partial pressure computation. Minor bugs (voltage string parsing, numpy truth-value ambiguity) were fixed through iterative testing.

Course Correction: CO vs CO2

> **Human:** For stage 3, change it to the CO mass fraction, not CO2, sorry.

A one-line fix – illustrating how the human directed the specific content while the AI handled implementation.

Course Correction: Undo Edits

> **Human:** Undo the edits please and resave the previous version.

When the human changed their mind about a modification to the plotting code, the AI reverted the changes cleanly.

---

6. Phase 5: Physicality Checks (March 26)

This phase exemplifies the core value of human-AI collaboration: the AI performed systematic numerical analysis while the human applied physical intuition to identify problems.

> **Human:** Perform physicality checks and debug plots. Examine the plots and their trends carefully and compare to the physicality checks and physical intuition given in the guide.

The AI examined all plots, extracted numerical values from binary cache files, and produced a detailed physicality report. Key findings:

Confirmed correct: - OH- highest at gas side (x=0), decreasing toward electrolyte - CO2(aq) depleted at x=0 by CO2R, higher toward electrolyte - pH gradient: 13.5 (gas side) to ~9 (electrolyte) - R_CO2R spikes at x=0 where CO2 is available - FE_CO peaks at intermediate voltage then declines (matching experiment qualitatively)

Flagged as expected model limitations: - Cs+ = 7.2 M at x=0 under high current – unphysically high but inherent to the model (no electromigration for spectator cations)

Human Challenges AI Interpretation

The human then probed specific features of the gas profiles:

> **Human:** The CO2 partial pressure increases across the domain for -0.3 V, how is that possible? And the H2 partial pressure curves have different shapes for different voltages, how is that possible?

The AI explained both phenomena with quantitative reasoning:

1. **CO2 increase:** The Sechenov salting-out effect at ~1M ionic strength reduces CO2 solubility by ~24%. The Stage 2 warm-start had dissolved CO2 set without Sechenov correction. When Stage 3 activates coupled phase transfer, the dissolved CO2 is above the Sechenov-corrected equilibrium, driving CO2 degassing from liquid to gas.
2. **Different H2 shapes:** Competition between diffusion (H2 from CL toward flow-field) and Darcy advection (direction depends on net gas mass source). At CO2R-dominated voltages, each mole of CO2 consumed (44 g/mol) produces only 1 mole CO (28 g/mol) – net 16 g/mol mass loss drives Darcy inward, opposing H2 diffusion. At HER-dominated voltages, this deficit weakens.

Human Identifies Real Problems

> **Human:** The gas phase results do not look correct. The partial pressures are expected to be roughly linear with respect to position and the inlet boundary conditions for the gases do not appear to be respected. The pCO2 should be constant around 0.75 atm at the inlet and the pCO and pH2 should be constant at 0 at the inlet. Also why do these graphs start at x=50?

This was a critical intervention. The human's physical intuition identified that: 1. Gas profiles should be approximately linear in the GDL 2. Inlet BC values (pCO2 ~ 0.75 atm, pCO = pH2 = 0) should be visible at x = -325 um 3. The x-axis was wrong – plots started at x ~ 50 um instead of x = -325 um

The AI investigated the coordinate system, initially misidentified the mesh orientation, self-corrected after reading the residual assembly code, and ultimately fixed the plotting to use Julia's physical coordinates [-325, +10] um with an explicit inlet face point at x = -325 um.

> **Human:** Change the plotting to not convert the x values from what Julia has, so plot from x = [-325, +10] um. If possible, add a point that shows the boundary condition at -325, so add the edge of the first cell not the center.

---

7. Phase 6: Code Audit (March 26–27)

> **Human:** Reexamine all Julia code and the implementation guide and double check all physicality checks, boundary conditions, and equations and ensure that everything is being properly implemented.

The AI launched three parallel audit agents examining (1) liquid residual assembly, (2) gas transport equations, and (3) kinetics and parameters – each cross-referencing the Julia code against the implementation guide equation by equation.

Findings

HIGH – GDL/CL Interface Flux Not Conserved

At the GDL/CL boundary, the same physical face was evaluated with different transport properties depending on which cell's residual was being computed. GDL permeability (8.4e-13 m^2) and CL permeability (8.0e-16 m^2) differ by ~1000x. The flux leaving the last GDL cell did not equal the flux entering the first CL cell, violating mass conservation.

MEDIUM – H2O Flux Closure

The H2O closure `J[4] = -(J[1]+J[2]+J[3])` forced zero total mass flux, which is only correct for pure diffusion. With Darcy advection, total mass flux should equal rho * u_G.

MEDIUM – CO2 Left BC Missing Sechenov

The dissolved CO2 Dirichlet BC at the flow-field face used `H_eff_base()` instead of `sechenov_H_eff()`. At ~1M ionic strength, this was a ~24% error.

---

8. Phase 7: Implementing Fixes (March 27)

> **Human:** Start with the two medium items.

The human directed the order of fixes, prioritizing the simpler ones first. After both MEDIUM fixes were applied:

> **Human:** Please do that and rerun it.

Stage 3 re-ran: 734/734 converged. The human then asked about the HIGH finding:

> **Human:** What was the high priority discrepancy found?

After the AI explained the interface flux conservation issue:

> **Human:** Okay try that.

The harmonic-mean fix was implemented:

```
D_eff_interface = 2.0 .* D_eff_GDL .* D_eff_CL ./ (D_eff_GDL .+ D_eff_CL .+ 1e-30)
kappa_interface = 2.0 * kappa_GDL * kappa_CL / (kappa_GDL + kappa_CL)
```

Stage 3 re-ran again: 734/734 converged. The impact was dramatic – gas profiles became nearly linear in the GDL (as the human had expected from the start), and the pressure drop reduced from ~7 kPa to ~0.1 kPa.

Pressure Convention Discussion

> **Human:** Should the delta p increase? Wouldn't there be pressure drop over the porous electrode?

The AI explained that the plot was already showing a pressure drop (p_inlet - p(x) > 0), but the convention was confusing. The human directed:

> **Human:** Define pressure drop as p - p_total and replot.

This made pressure drops appear as negative values, which is more intuitive – gas pressure decreases as it flows from the inlet into the electrode.

---

9. Phase 8: Parameter Fitting (March 27–28)

With all code fixes verified, the model still overpredicted CO2R current by 3–12x. The human directed fitting:

> **Human:** Yes go ahead with parameter fitting.

The AI wrote `fit_params.jl`, a fitting script that:

1. Loaded warm-starts from Stage 3 cache at the 5 experimental voltages
2. For each candidate parameter set (i0_CO2R, lambda_MHC), re-solved the coupled system at those voltages
3. Minimized weighted log-space residuals: CO current weighted 2x over H2
4. Used grid search (42 evaluations) followed by Nelder-Mead refinement (19 iterations)

The fitting leveraged Julia's `Ref`-backed parameter system, which had been designed for exactly this purpose – parameters could be changed at runtime without recompilation.

Results

Parameter	Initial	Fitted	Change
i0_CO2R [A/m^2]	540.0	51.13	10.6x reduction
lambda_MHC [eV]	1.800	1.5229	15% reduction

Comparison at Experimental Voltages (Post-Fit)

V_RHE [V]	i_CO model	i_CO exp	i_H2 model	i_H2 exp
-0.816	37.0	35.1	54.5	29.9
-1.205	204.4	189.6	89.4	49.3
-1.644	219.9	299.1	244.0	119.6
-2.079	172.1	204.3	636.6	449.9
-2.413	116.8	64.6	1357.8	850.5

CO2R current improved from 3–12x overprediction to within 2x at all voltages. HER remains overpredicted at high overpotential – a separate fitting target identified for future work.

> **Human:** Yes apply the fit and rerun.

Final Stage 3 run (run8): 734/734 converged in ~15 minutes. Stage 4 re-run: 734/734 in ~30 minutes. All plots regenerated.

---

10. Session Recovery

A notable aspect of this collaboration was the need to recover from an AI session crash. Mid-way through the code audit phase, the Claude Code session crashed, losing all conversation context. In the subsequent session:

> **Human:** You crashed in another window. What is the most recent prompt from me you have in your memory?
>
> **AI:** Based on what I have access to, my memory system only stores high-level project context, not conversation transcripts or individual prompts.

The human then pasted a recovered transcript from the crashed session, enabling the AI to reconstruct the full context and continue from where the work was interrupted:

> **Human:** Any chance you can read this to get context? It's all I could copy paste from the crashed window. Please start saving our conversation so there is a complete record of the collaborative build process.

This incident prompted the AI to implement more thorough project memory, saving build history, user profile, and explicit instructions to maintain records across sessions.

---

11. Reflections on Human-AI Collaboration

Division of Labor

The collaboration exhibited a natural division of labor:

Human contributions: - Domain expertise and physical intuition (e.g., "partial pressures should be roughly linear," "the inlet BCs aren't being respected") - Quality control and validation ("perform physicality checks") - Direction-setting ("start with the two medium items," "yes apply the fit and rerun") - Course corrections ("undo the edits," "change it to CO mass fraction not CO2") - Identifying when results don't match expectations

AI contributions: - Code generation from specifications (13 Julia files, 1 Python file) - Systematic debugging (tracing sign errors through DOF indices) - Numerical analysis (extracting and interpreting binary solution data) - Parallel code auditing (3 simultaneous audit agents) - Automated fitting (grid search + Nelder-Mead) - Real-time monitoring of long-running computations

Patterns Observed

1. **Specification-driven implementation:** The human provided a detailed implementation guide; the AI translated it to code. Bugs arose from translation errors (sign conventions, index offsets), not from misunderstanding the physics.
2. **Human as validator:** The human rarely wrote code directly but consistently identified when outputs didn't match physical expectations. Each such identification led to concrete fixes.
3. **Iterative refinement:** The build cycle was: implement -> run -> examine -> identify issue -> fix -> re-run. Each cycle took 15–45 minutes (dominated by solver runtime and network I/O).
4. **AI self-correction:** In one instance (coordinate system), the AI initially misidentified the mesh orientation, then self-corrected after reading the assembly code more carefully. The human's insistence on physical correctness ("the graphs start at x=50, why?") was the trigger.
5. **Crash recovery:** The AI's persistent memory system enabled continuation across sessions, though the crashed session required manual transcript recovery by the human.

---

12. Summary of All Changes

Bugs Fixed (5)

1. Filename parse off-by-one (all stage files)
2. O(n^2) network I/O (all stage files)
3. Wrong voltage sweep order (Stages 2, 3)
4. **Neumann BC sign error** (assembly.jl, run_stage3.jl) – critical, caused 100% Stage 2 failure
5. FD Jacobian dead zone at clamp boundary (solver.jl)

Code Audit Fixes (3)

6. Sechenov correction at CO2 inlet BC (run_stage3.jl)
7. H2O flux closure accounting for Darcy mass flux (gas_transport.jl)
8. **Harmonic-mean transport at GDL/CL interface** (run_stage3.jl) – restored flux conservation, linearized gas profiles

Parameter Calibration

9. MHC kinetic parameters fitted: i0_CO2R 540 -> 51.13 A/m^2, lambda_MHC 1.80 -> 1.52 eV

Visualization

10. 14+ plot types implemented in plot_results.py (595 lines)
11. Coordinate system corrected to physical coordinates [-325, +10] um
12. Pressure convention changed to p - p_total

---

13. File Inventory

```
co2r_bulk/
  params.jl            Physical constants, fitted kinetic parameters
  mesh.jl              CL/GDL mesh construction
  chemistry.jl         Buffer equilibrium system (5 species)
  kinetics.jl          MHC CO2R, BV HER, phase transfer, Sechenov
  transport.jl         Liquid diffusion, effective diffusivities
  gas_transport.jl     SM flux, mixture-average D, FSG, Darcy
  assembly.jl          Liquid FV residual, diagnostics
  solver.jl            Dense FD Newton, row scaling, PAC continuation
  run_stage1.jl        Stage 1 driver (Dirichlet BCs)
  run_stage2.jl        Stage 2 driver (Neumann BC)
  run_stage3.jl        Stage 3 driver (coupled gas+liquid)
  run_stage4.jl        Stage 4 driver (pressure diagnostics)
  fit_params.jl        Parameter fitting (grid + Nelder-Mead)
  plot_results.py      All visualization
  output_stage1/       734 cached solutions, CSV, plots
  output_stage2/       734 cached solutions, CSV, plots
  output_stage3/       734 cached solutions, CSV, plots
  output_stage4/       734 cached solutions, CSV, plots
```

Document generated March 29, 2026. Record of collaborative work between a human researcher and Claude Opus 4.6 (Anthropic) via Claude Code.